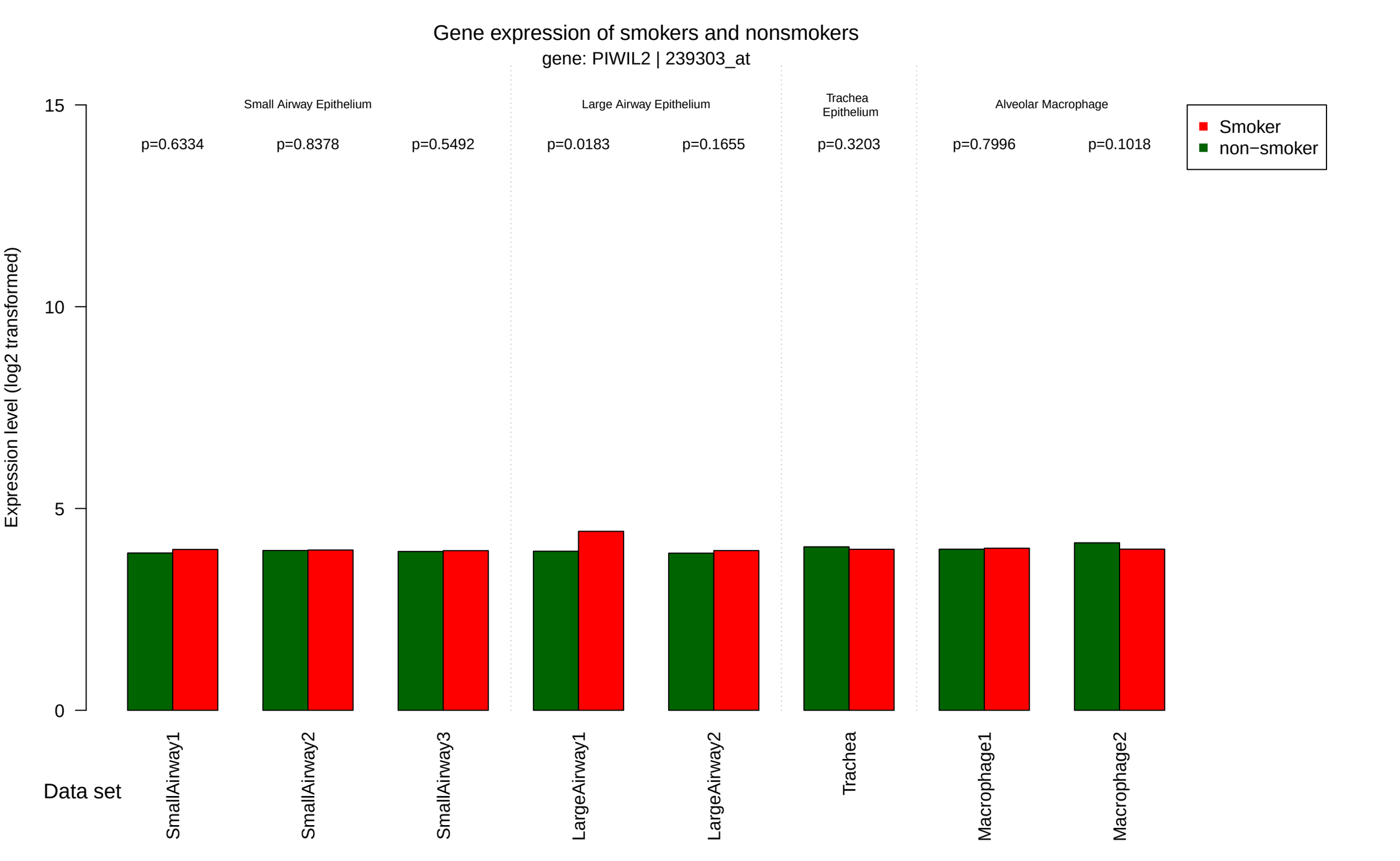
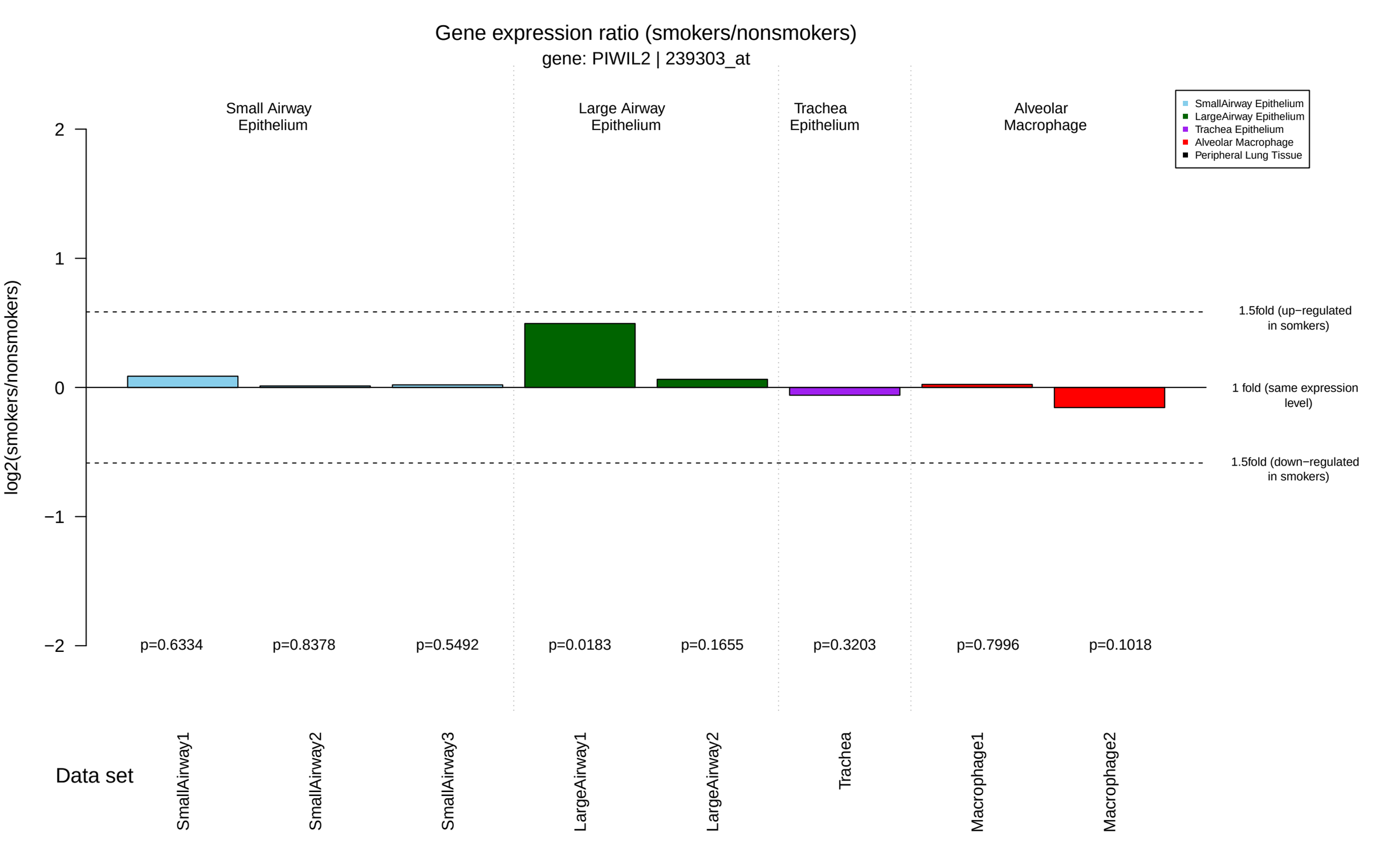
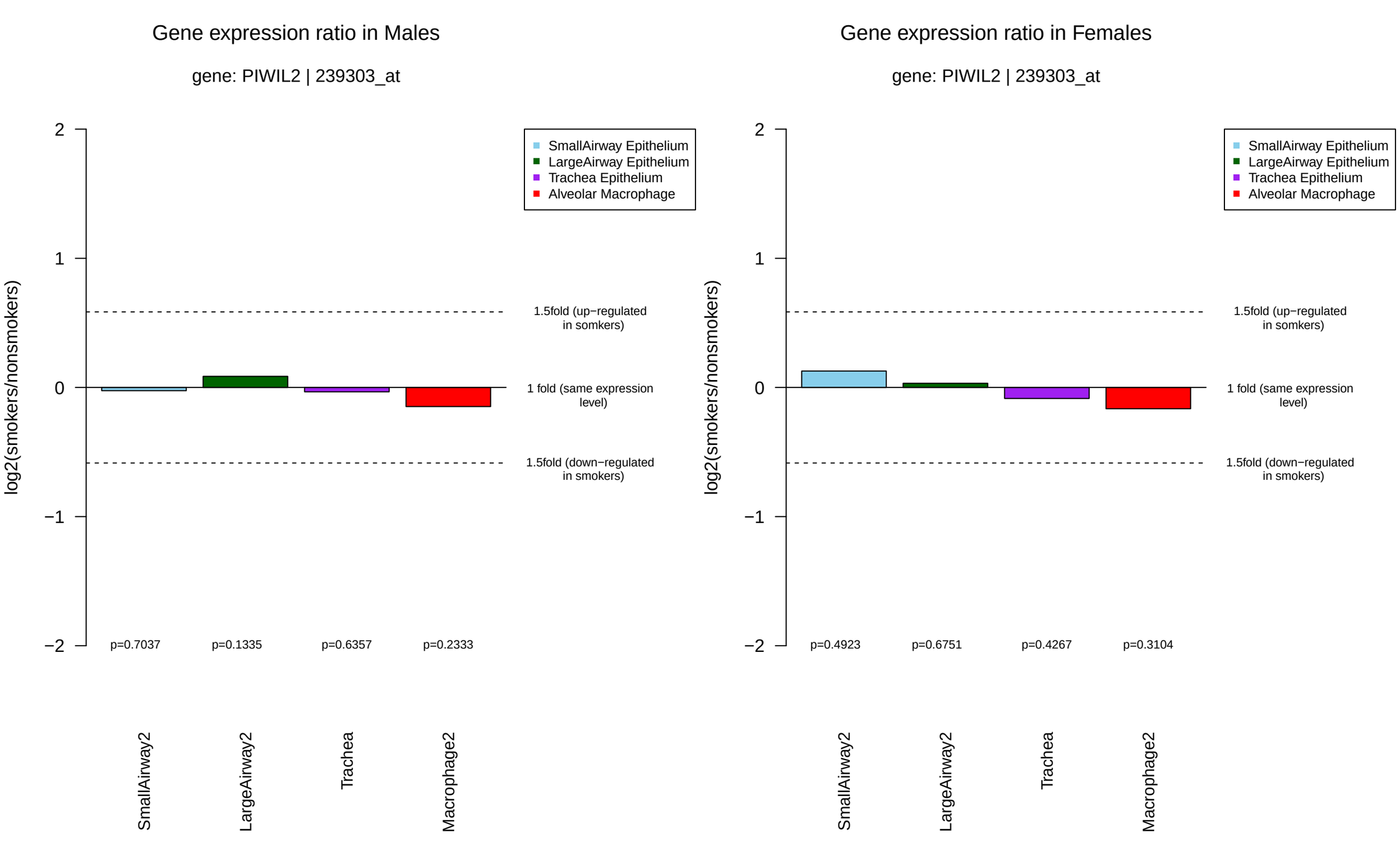
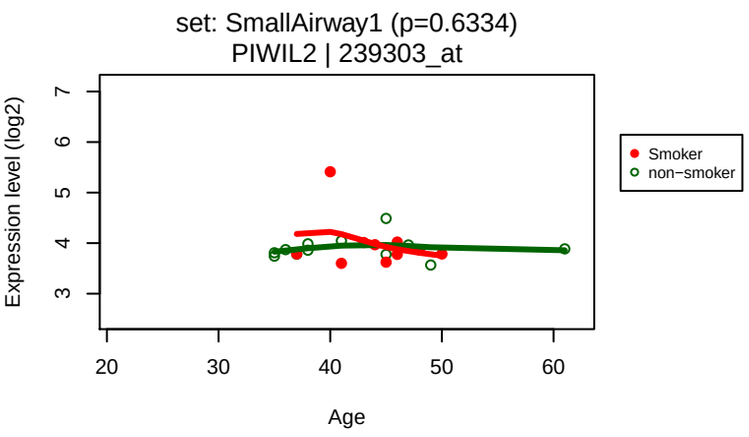
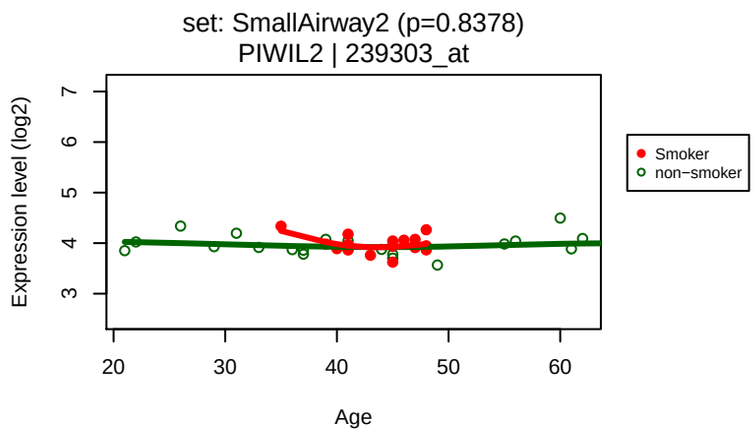
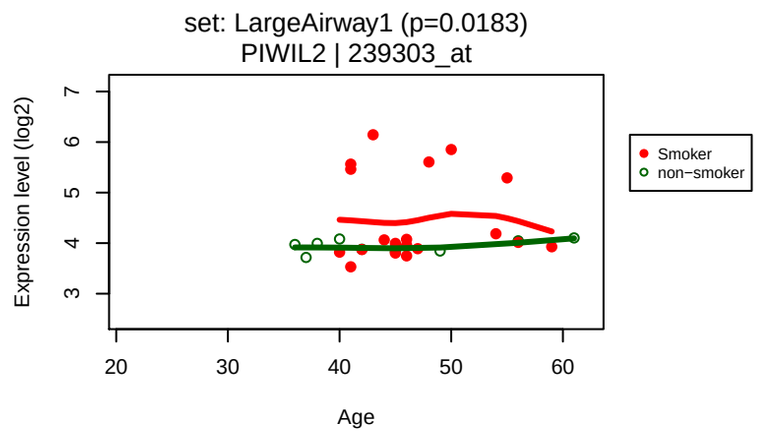

Gene expression of smokers and nonsmokers
gene: PIWIL2 | 239303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6334
p=0.8378
p=0.5492
p=0.0183
p=0.1655
p=0.3203
p=0.7996
p=0.1018
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

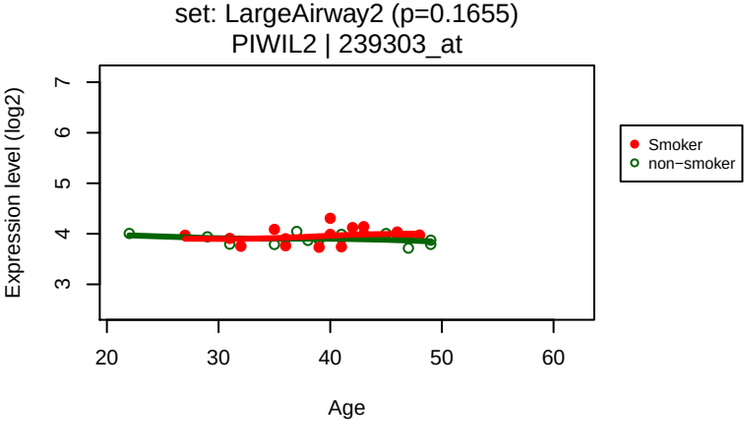

Gene expression ratio (smokers/nonsmokers)
gene: PIWIL2 | 239303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6334
p=0.8378
p=0.5492
p=0.0183
p=0.1655
p=0.3203
p=0.7996
p=0.1018
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

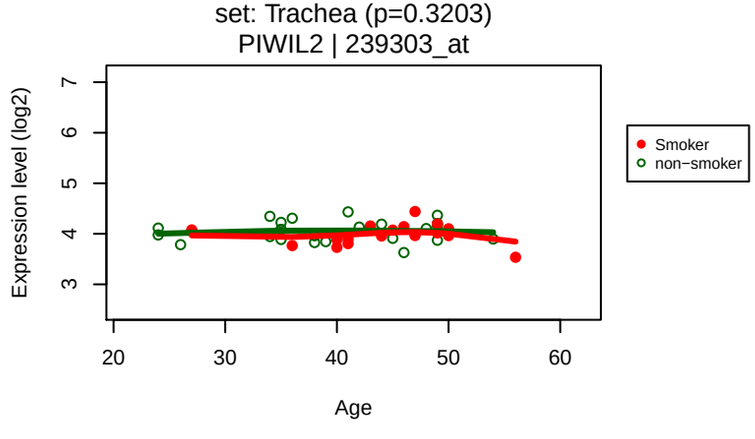

Gene expression ratio in Males
gene: PIWIL2 | 239303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7037
p=0.1335
p=0.6357
p=0.2333
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIWIL2 | 239303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4923
p=0.6751
p=0.4267
p=0.3104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

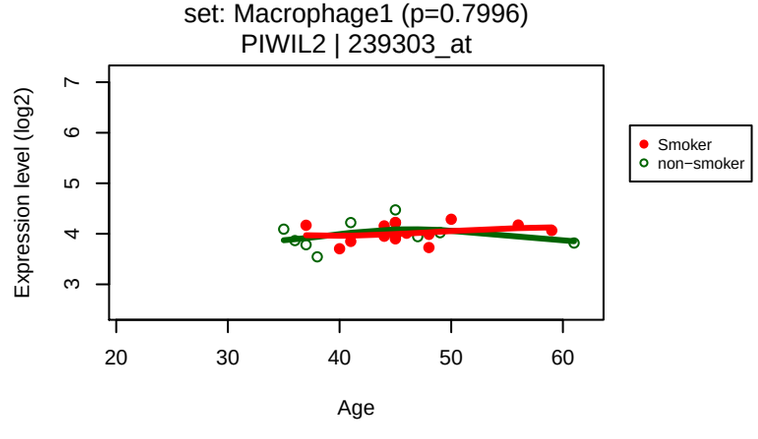

set: SmallAirway1 (p=0.6334)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker

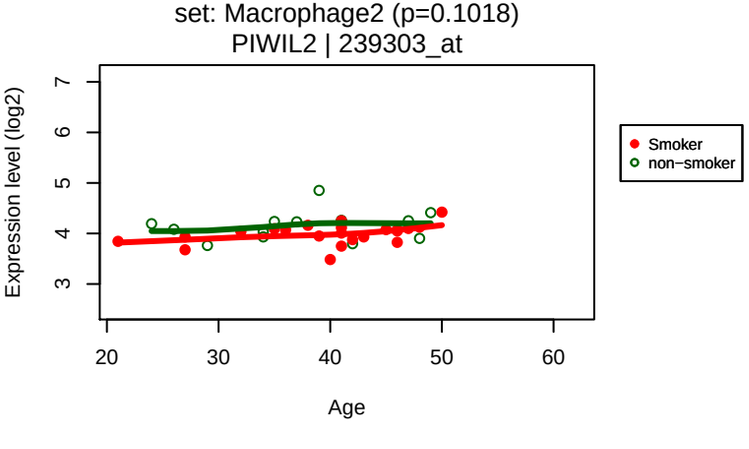

set: SmallAirway2 (p=0.8378)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker

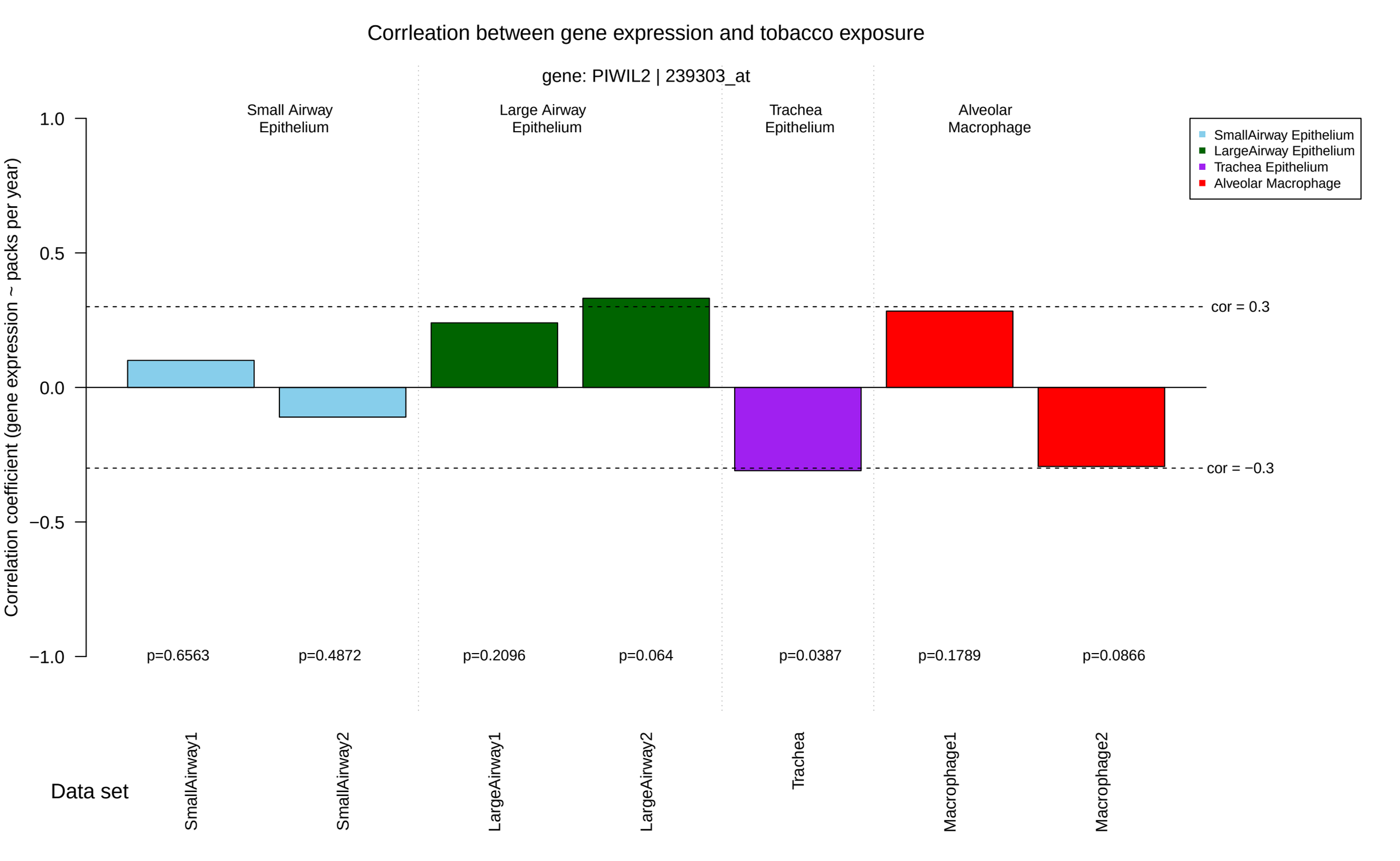

set: LargeAirway1 (p=0.0183)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1655)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3203)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7996)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1018)
PIWIL2 | 239303_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIWIL2 | 239303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6563
p=0.4872
p=0.2096
p=0.064
p=0.0387
p=0.1789
p=0.0866
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2